Experiment 5
Relational and Multidimensional Reporting Using Microsoft Power BI

Aim: To generate relational and multidimensional reports on the star schema built in Experiment 3 and extended in Experiment 4, using table and matrix visuals, the Geo_Hierarchy, and comparative charts in Power BI, and to combine them into an interactive dashboard that supports business decision-making.

OBJECTIVES:

- To generate relational reports using Table visuals in Microsoft Power BI.
- To create multidimensional reports using Matrix visuals and the existing star schema.
- To implement the `Geo_Hierarchy` (Region → State → City) and perform Drill-Down and Drill-Up operations.
- To apply sorting and filtering for detailed business data analysis.
- To create comparative charts for Sales by Category, Profit by Region, Top 10 Customers by Sales, and Year-wise Sales Trends.
- To develop an interactive dashboard using KPI cards and slicers for Region, Category, and Year.
- To analyze business performance using Sales, Profit, and Quantity and support data-driven decision-making.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

Microsoft Power BI is a Business Intelligence tool used to transform raw data into meaningful information through reports, charts, and interactive dashboards. In this experiment, the Global Superstore dataset is used to generate relational and multidimensional reports based on the star schema developed in previous experiments.

Relational reports use Table visuals to display detailed information such as Order ID, Customer Name, Product Name, Region, Sales, and Profit. Sorting and filtering help organize records and analyze specific business transactions. Multidimensional reports use Matrix visuals to summarize Sales, Profit, and Quantity across different geographical levels.

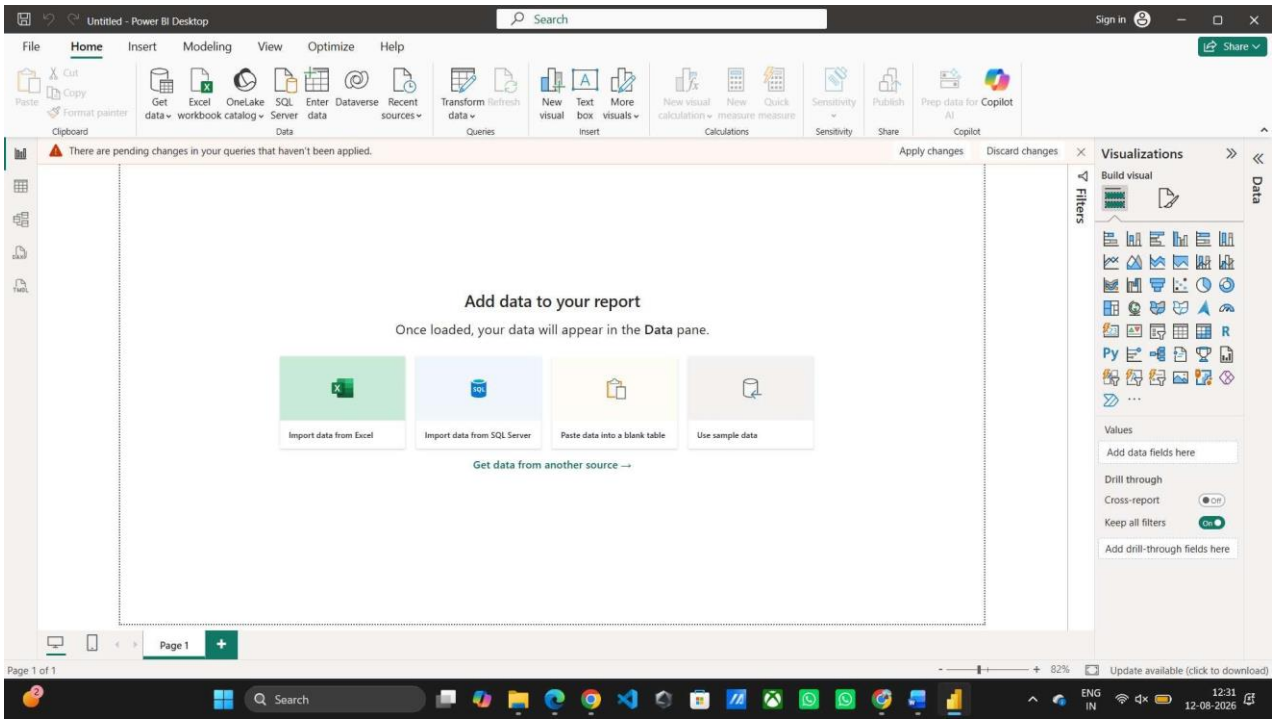
The `Geo_Hierarchy` (Region → State → City) enables Drill-Down and Drill-Up operations for exploring data at different levels of detail. Comparative charts analyze Sales by Category, Profit by Region, Top 10 Customers by Sales, and Year-wise Sales Trends. Finally, KPI cards and interactive slicers make the dashboard dynamic and help users compare business performance and support data-driven decision-making.

PROCEDURE:

Part A: Generate Relational Report

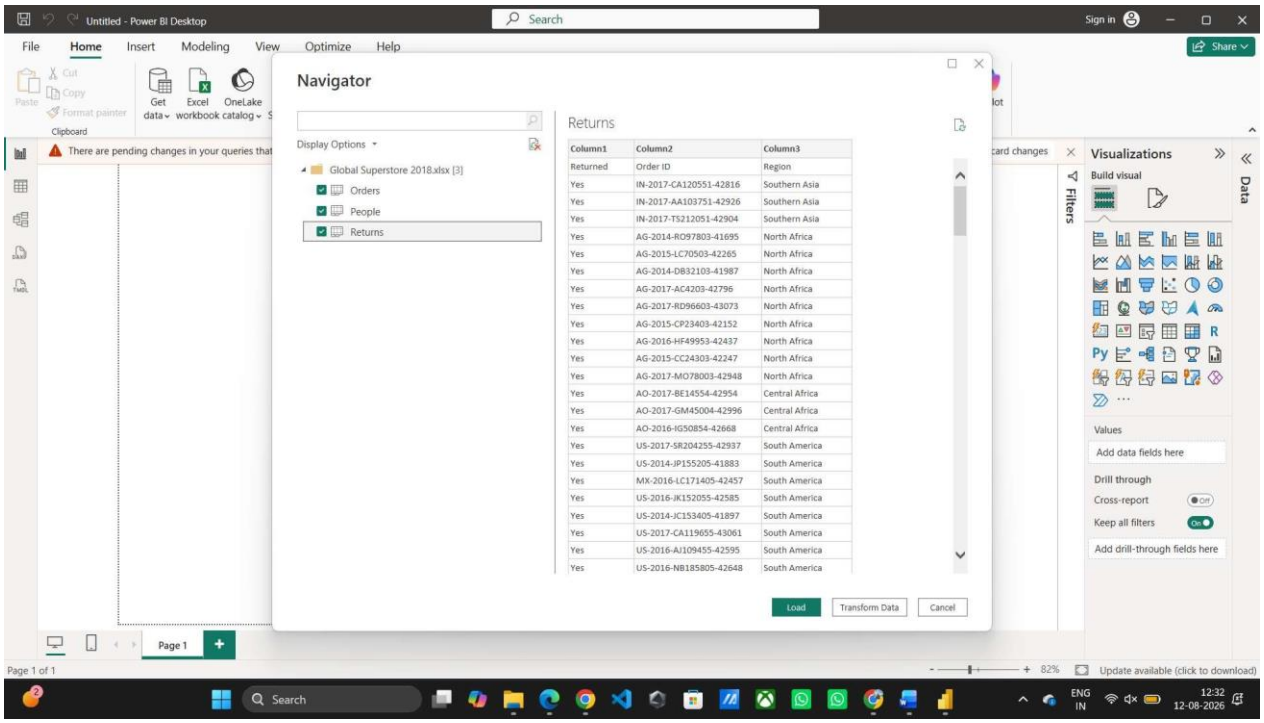
Step 1: Open the Power BI project.

Launch Microsoft Power BI Desktop and open the project created in the previous experiments using the Global Superstore dataset.



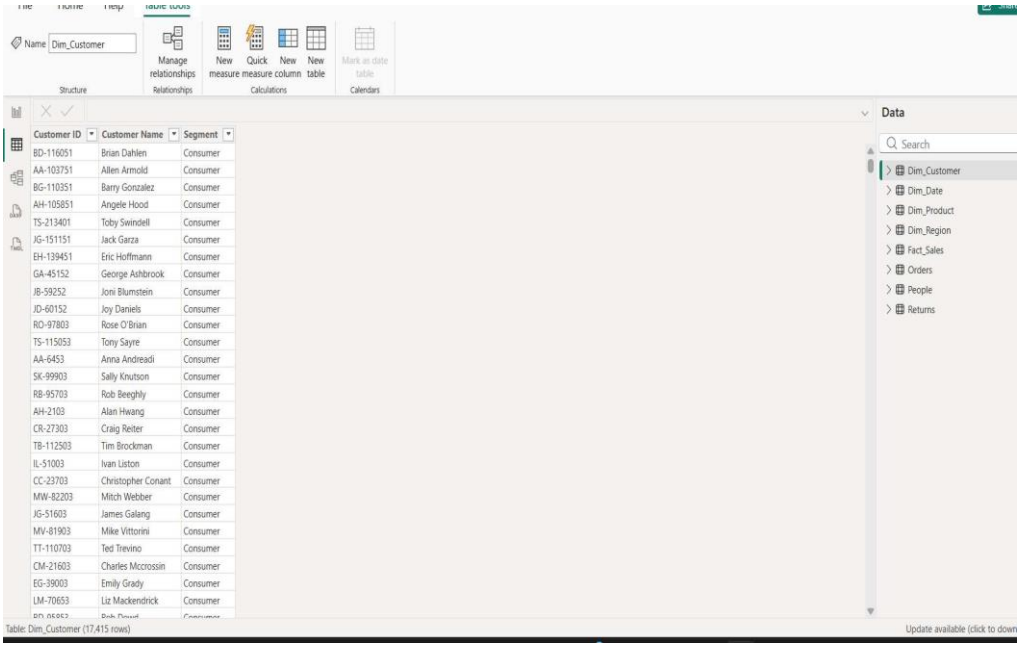
Step 2: Verify the imported tables.

Check the Data/Fields pane and verify that the existing tables, such as `Orders`, `Fact_Sales`, `Dim_Product`, `Dim_Region`, and `Dim_Date`, are available.



Step 3: Create a Table Visual.

1. Open Report View.
2. Select an empty area of the report canvas.
3. Choose the Table visual from the Visualizations pane.



Step 4: Add the required fields.

Add the following fields to the Table visual:

- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit

Select the appropriate fields from the existing tables in the model.

Step 5: Apply sorting.

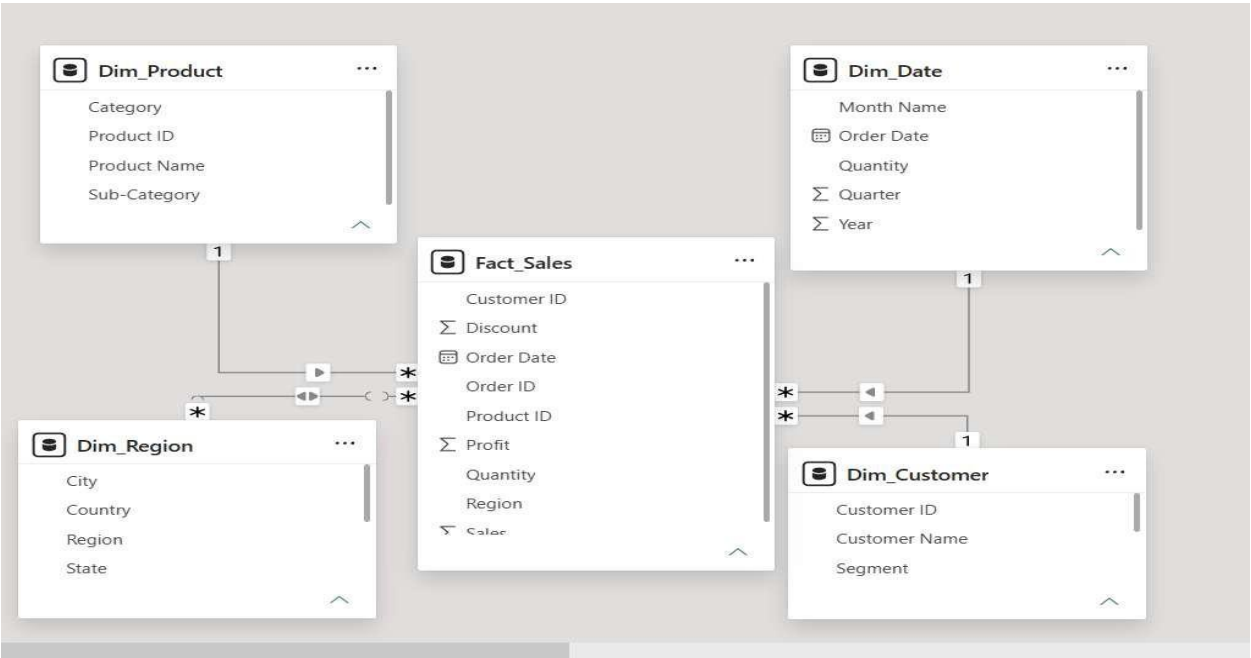
Sort the table by `Sales` in descending order to display high-sales transactions first. Repeat the operation using `Profit` to arrange records according to profitability.

Step 6: Apply filtering.

Use the Filters pane to filter records based on `Region`. For example, select `West` to display transactions belonging to that region. Apply numerical filters such as `Sales greater than 1000` or `Profit greater than 0` to analyze specific records.

Step 7: Export the report.

Select the Table visual, click the three-dot menu, and choose Export data. Save the available exported file and open it in Microsoft Excel to examine the transactional records.



Step 8: Record observations.

Observe how sorting and filtering help organize records, identify high-value transactions, and analyze specific regions or profitable sales.

Part B: Generate Multidimensional Report

Step 9: Verify the geographic hierarchy.

In the Data/Fields pane, locate the existing `Geo_Hierarchy`. It should contain the following levels:

`Region → State → City`

If the hierarchy does not exist, create a new hierarchy using these fields in the specified order.

Step 10: Insert a Matrix Visual.

Select an empty area of the report canvas and choose the Matrix visual from the Visualizations pane.

Step 11: Add the hierarchy to Rows.

Drag `Geo_Hierarchy` into the Rows section of the Matrix visual.

Step 12: Add measures to Values.

Add the following numerical fields to Values:

- Sales — Sum
- Profit — Sum
- Quantity — Sum

The Matrix will summarize business performance at the geographic level.

Step 13: Perform Drill-Down.

Use the drill controls on the Matrix or a separate chart containing the hierarchy to move from Region to State and then to City. Examine the aggregated Sales, Profit, and Quantity at each level.

Step 14: Perform Drill-Up.

Use the Drill Up control to return from City to State and then to Region. Compare the summarized values at each geographic level.

Step 15: Record observations.

Identify the regions, states, and cities with comparatively higher sales and profit. Observe how aggregated values change as the analysis moves between geographic levels.

Part C: Generate Comparative Analysis Report

Step 16: Create Sales by Category.

Insert a column chart or bar chart. Place `Category` on the category axis and `Sales` on the value axis, using Sum aggregation. Title the visual Sales by Category.

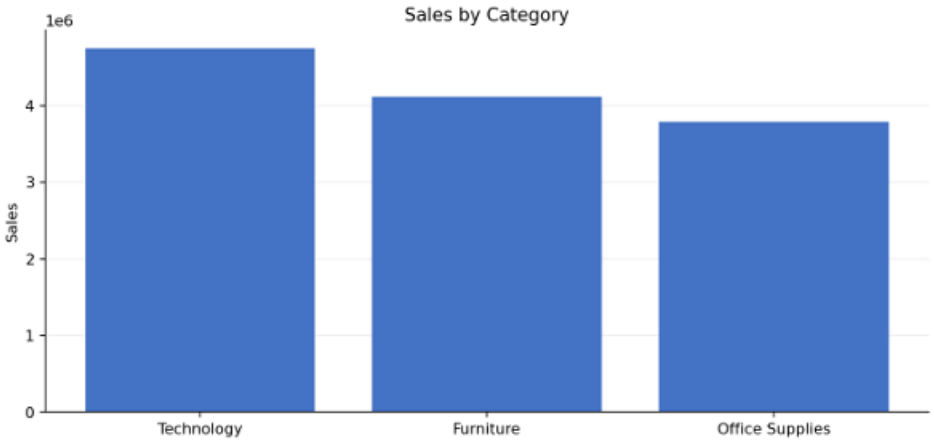
Step 17: Create Profit by Region.

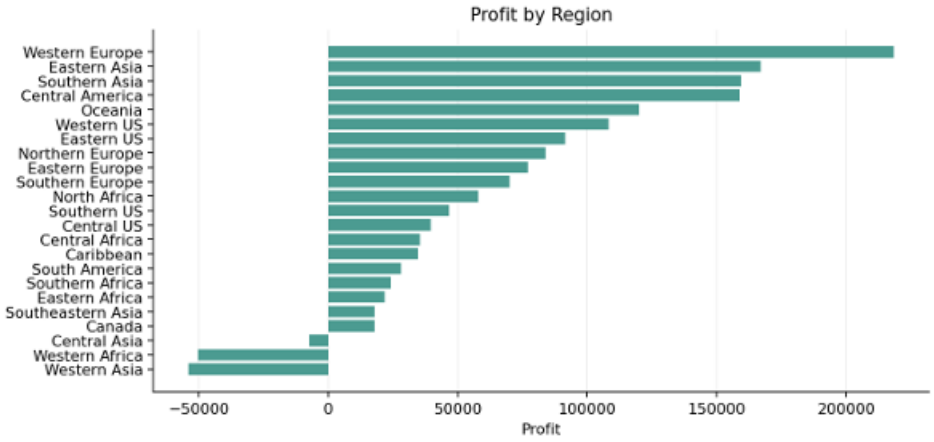
Insert a clustered bar chart. Place `Region` on the category axis and `Profit` on the value axis, using Sum aggregation. Sort the chart by profit to compare regional profitability. Title it Profit by Region.

Step 18: Create Top 10 Customers by Sales.

Insert a bar chart and add `Customer Name` and `Sales`. In the visual-level Filters pane, select `Customer Name`, choose the Top N filter type, enter `10`, add `Sales` under By value, and apply the filter.

Title the visual Top 10 Customers by Sales.


Sales by Category
1e6
Sales
0
1
2
3
4
Technology
Furniture
Office Supplies


Profit by Region
Western Europe
Eastern Asia
Southern Asia
Central America
Oceania
Western US
Eastern US
Northern Europe
Eastern Europe
Southern Europe
North Africa
Southern US
Central US
Central Africa
Caribbean
South America
Southern Africa
Eastern Africa
Southeastern Asia
Canada
Central Asia
Western Africa
Western Asia
−50000
0
50000
100000
150000
200000
Profit

Step 19: Create the Year-wise Sales Trend.

Insert a Line Chart. Use `Year` from `Dim_Date` on the X-axis and Total Sales on the Y-axis. Arrange the years in ascending order and title the visual Year-wise Sales Trend.

Step 20: Compare business performance.

Analyze the resulting charts to compare category-wise sales, region-wise profit, customer contributions, and changes in sales across years. Record the major observations from each visual.

Part D: Create an Interactive Dashboard

Step 21: Create KPI Cards for Total Sales.

Insert a Card visual and add the existing `Total Sales` measure, or create the following DAX measure:

```
Total Sales = SUM(Fact_Sales[Sales])
```

Set the visual title to Total Sales.

Step 22: Create a Total Profit KPI.

Insert another Card visual and create or use the following measure:

```
Total Profit = SUM(Fact_Sales[Profit])
```

Set the visual title to Total Profit.

Step 23: Create a Total Orders KPI.

Insert another Card visual and create the following measure:

Total Orders =
DISTINCTCOUNT(Fact_Sales[Order ID])

Set the visual title to Total Orders. Distinct Count ensures that repeated order IDs across multiple product lines are not counted as separate orders.

Step 24: Add a Region slicer.

Insert a Slicer visual and add the `Region` field from the appropriate dimension table. This enables users to view business performance for a selected region.

Step 25: Add a Category slicer.

Insert another Slicer visual and add `Category` from `Dim_Product`. This enables users to filter the dashboard by product category.

Step 26: Add a Year slicer.

Insert another Slicer visual and add `Year` from `Dim_Date`. This allows users to view business performance for a selected year.

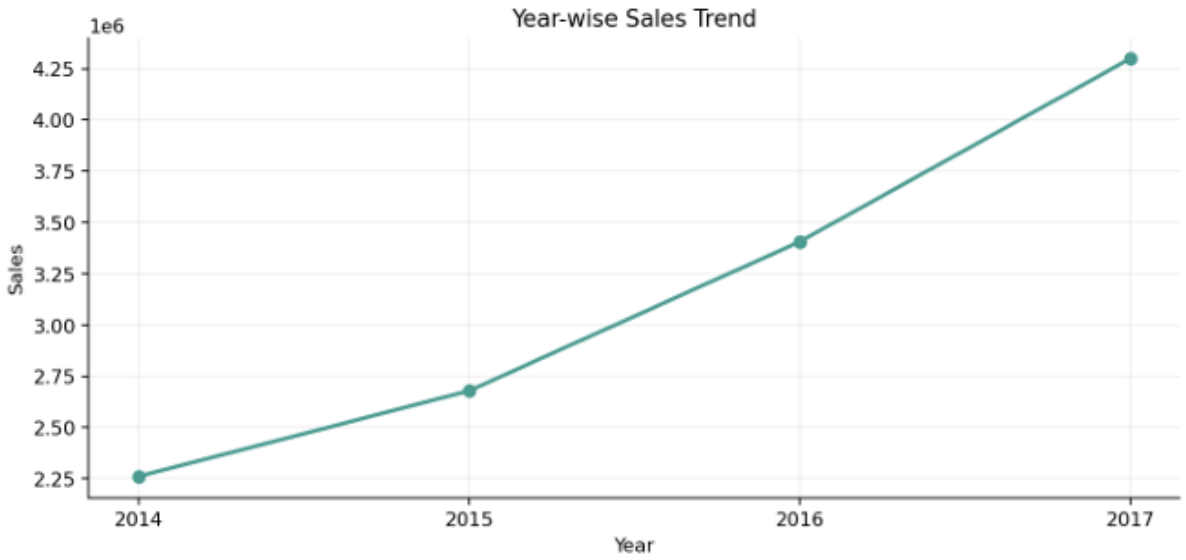
Region	State	City	Sales	Profit	Quantity
Eastern US	New York	New York City	$256,368	$62,037	3,417
Western US	California	Los Angeles	$175,851	$30,441	2,879
Southeastern Asia	National Capital Region	Manila	$120,887	$-11,159	1,661
Western US	Washington	Seattle	$119,541	$29,156	1,590
Western US	California	San Francisco	$112,669	$17,507	1,935
Eastern US	Pennsylvania	Philadelphia	$109,077	$-13,838	1,981
Oceania	New South Wales	Sydney	$101,946	$16,003	975
Southeastern Asia	Jakarta	Jakarta	$94,321	$3,827	1,226
Central America	Distrito Federal	Mexico City	$85,700	$13,336	1,138
Central America	Managua	Managua	$83,707	$17,854	1,233
Northern Europe	England	London	$80,764	$15,605	976
Caribbean	Santo Domingo	Santo Domingo	$78,714	$-3,570	1,649
Southeastern Asia	Bangkok	Bangkok	$76,050	$-7,125	1,083
Oceania	Queensland	Brisbane	$75,729	$10,526	689
Oceania	Victoria	Melbourne	$73,844	$11,295	811
Oceania	Queensland	Gold Coast	$72,627	$4,454	728
Central US	Texas	Houston	$64,505	$-10,154	1,466
Oceania	Western Australia	Perth	$64,292	$8,634	737

Step 27: Arrange the dashboard layout.

Organize the KPI cards at the top of the report. Place the slicers below the title or along the side, and arrange Sales by Category, Profit by Region, Top 10 Customers by Sales, and Year-wise Sales Trend in a clear layout.

Step 28: Verify slicer interactions.

Select different values in the Region, Category, and Year slicers. Verify that the KPI cards and relevant charts update according to the selected filters. If a visual does not respond as expected, check Format → Edit interactions and verify the appropriate filter interaction.



Step 29: Record observations.

Observe how the interactive dashboard supports business analysis by allowing users to compare sales, profit, order counts, customer performance, and yearly trends under different filter conditions.

Part E: Validate and Compare Reports

Step 30: Compare relational and multidimensional reports.

Compare the detailed transaction records in the Table visual with the aggregated results in the Matrix visual. Identify how each approach supports a different level of business analysis.

Step 31: Validate hierarchical operations.

Verify that the `Geo_Hierarchy` supports Drill-Down from Region to State and City and Drill-Up from City to State and Region. Capture screenshots showing the different hierarchy levels.

Result

Relational and multidimensional reports were generated using Microsoft Power BI and the Global Superstore dataset. Sorting and filtering were applied to transactional data, while Matrix visuals and the `Geo_Hierarchy` were used for aggregated geographic analysis. Comparative charts, Drill-Down and Drill-Up operations, KPI cards, and interactive slicers were used to analyze business performance and develop an interactive dashboard for decision-making.

Step 32: Summarize the findings.

Prepare a summary of sorting, filtering, aggregation, hierarchical analysis, comparative visualizations, and interactive dashboard operations. Save the Power BI project and capture screenshots of the completed reports for documentation.

LEARNING OUTCOMES:

- Learned to generate relational and multidimensional reports using Microsoft Power BI.
- Understood sorting and filtering techniques for analyzing business data.
- Learned to create and use hierarchical structures for Drill-Down and Drill-Up operations.
- Gained knowledge of DAX measures for calculating Total Sales, Total Profit, Total Orders, and Profit Margin.
- Developed comparative visualizations to analyze category-wise sales, regional profit, customer performance, and yearly sales trends.
- Learned to create interactive dashboards using KPI cards and slicers.
- Understood how Business Intelligence supports data analysis and business decision-making.